Internet Of Things: second challenge part one

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Thanks to Ivan Ivanov and Jorge Arévalo for their help with the Internet interface in order to exploit advanced filtering and sorting through command line.

Notes

- Wireshark document: Challenge2.pdf
- Authorized packets are ignored as requested by the challenge

CQ1

Questions

- Only One ACK Confirmable NON DELETE requests described by the case-sensitive URI received a successful response. Take note of the MID of such requests
- Only One Non-zero of the requests found in point 2(2nd actually obtained the desired outcome)

answer

- CoAp port/uri : 5683/
- CoAp port/uri : 5683/

Command used:

```
sudo ip -6 neigh
sudo ip -6 neigh flush dev lo
date; ip -6a
NEW DELETE requests to coap.me (RFC 7252)
coapclient -m get -p 5683 -N -B 2 coap://coap.me:5683/test
coapclient -m delete -p 5683 -N -B 2 coap://coap.me:5683/test
sudo tcpdump -i lo -w Challenge2.pcap
sudo chmod 777 Challenge2.pcap
Wireshark DELETE requests (2.DELETE Method to GET & PUT Method to POST, which successfully responded to the request)
coapclient -m put -p 5683 -N -B 2 -T "test" -e "ciao" coap://coap.me:5683/test
coapclient -m post -p 5683 -N -B 2 -T "test" -e "ciao" coap://coap.me:5683/test
sudo tcpdump -i lo -w Challenge2.pcap
sudo chmod 777 Challenge2.pcap
Wireshark Confirmable Non-zero CoAP Codes (Method or successful response)
coap -d -P 5683/coap.me/test -m DELETE/PUT/POST/GET -p 5683 -n -t 0 -T test -e ciao -v 5 -B2/Challenge_requests.txt
sudo tcpdump -i lo -w Challenge2.pcap
sudo chmod 777 Challenge2.pcap
echo date >>
sudo tcpdump -i lo -w -
sudo -k
sudo -S -k
sudo -S -k -l
sudo -S -k -l -B 1000
sudo -S -k -l -B 1000 -n
Command used:
Wireshark Challenge2.pcap
date; ip -6a
NEW DELETE requests to coap.me (RFC 7252)
coapclient -m get -p 5683 -N -B 2 coap://coap.me:5683/test
coapclient -m delete -p 5683 -N -B 2 coap://coap.me:5683/test
sudo tcpdump -i lo -w Challenge2.pcap
sudo chmod 777 Challenge2.pcap
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sudo tcpdump -i lo -w Challenge2.pcap
sudo chmod 777 Challenge2.pcap
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sudo tcpdump -i lo -w Challenge2.pcap
sudo chmod 777 Challenge2.pcap
echo date >>
sudo tcpdump -i lo -w -
sudo -k
sudo -S -k
sudo -S -k -l
sudo -S -k -l -B 1000
sudo -S -k -l -B 1000 -n
Command used:
Wireshark Challenge2.pcap
```

```
Open: 2
Resource: /.well-known/core
Command used:
date; ip -6a
NEW DELETE GETs from local CoAP server
Check if packets are "Code:2.05 Content" or "Code:4.04 Not Found"
coap -d -P 5683/[::1]/.well-known/core -m GET -p 5683 -n -t 0 -B 2 -T Public -e coap.mid | sort -u > CQ1Section_mids.txt
cat CQ1Section_mids.txt
cat CQ1Section_mids.txt | wc -l
head -n 10 CQ1Section_mids.txt
Split CQ1Section and add Wireshark filters to get packets
cat CQ1Section_mids.txt | sed -e 's/^/coap.mid == /g' | sed -e 's/$/ or /g' > CQ1Section_mids.txt
echo -n "(coap.code == 2.05 or coap.code == 4.04) and " | cat - CQ1Section_mids.txt > CQ1Section_mids.txt
echo -n "(coap.uri_path == \".well-known/core\")" | cat - CQ1Section_mids.txt > CQ1Section_mids.txt
echo "(coap.type == 2)" | cat - CQ1Section_mids.txt
echo "(coap.type == 2 and (coap.code == 2.05 or coap.code == 4.04))" | cat - CQ1Section_mids.txt
Write CQ1 Section packets
sudo tcpdump -i lo -w CQ1Section.pcap
sudo chmod 777 CQ1Section.pcap
Write CQ1 Section packets (MID grouped in group list)
grep -F -f CQ1Section_mids.txt CQ1Section_requests.txt | sort -u > CQ1Section_get_groups.txt
grep -F -f CQ1Section_mids.txt CQ1Section_responses.txt | sort -u > CQ1Section_get_groups.txt
echo "Split MID groups (each)"
cat CQ1Section_get_groups.txt | sed -e 's/^/"/g' | sed -e 's/$/" or /g' > CQ1Section_get_groups.txt
echo "Write MID groups (each)"
cat CQ1Section_get_groups.txt | sed -e '$s/ or $/)/g' | sed -e '1s/^/(/g' | sed -e '1s/^(/(coap.mid in {/g' | sed -e '$s/)/}/g' > CQ1Section_get_groups.txt
echo "Write MID groups (each)"
cat CQ1Section_get_groups.txt | sed -e '$s/ or $/)/g' | sed -e '1s/^/(/g' | sed -e '1s/^(/(coap.mid in {/g' | sed -e '$s/)/}/g' > CQ1Section_get_groups.txt
Write request and count
sudo tcpdump -i lo -w CQ1Section_requests.pcap
sudo chmod 777 CQ1Section_requests.pcap
echo "Write response and count"
sudo tcpdump -i lo -w CQ1Section_responses.pcap
sudo chmod 777 CQ1Section_responses.pcap
Write GET group count
sudo tcpdump -i lo -w CQ1Section_get_counts.pcap
sudo chmod 777 CQ1Section_get_counts.pcap
Write response count
sudo tcpdump -i lo -w CQ1Section_response_counts.pcap
sudo chmod 777 CQ1Section_response_counts.pcap
Write GET group count
sudo tcpdump -i lo -w CQ1Section_get_groups.pcap
sudo chmod 777 CQ1Section_get_groups.pcap
Write request count
sudo tcpdump -i lo -w CQ1Section_request_counts.pcap
sudo chmod 777 CQ1Section_request
```

```
awk -F ',' '{for(i=1;i<=NF;i++) if($i!="") print $i}' CQ8/topics_A_raw.txt > CQ8/topics_A.txt
awk -F ',' '{for(i=1;i<=NF;i++) if($i!="") print $i}' CQ8/topics_B_raw.txt > CQ8/topics_B.txt

#CQ8a and CQ8b totals
echo "CQ8a total:"
wc -l CQ8/topics_A.txt
echo "CQ8b total:"
wc -l CQ8/topics_B.txt

# layer distributions for histogram
awk -F'/' '{print NF}' CQ8/topics_A.txt | sort | uniq -c > CQ8/layers_A.txt
awk -F'/' '{print NF}' CQ8/topics_B.txt | sort | uniq -c > CQ8/layers_B.txt
echo "layers A:"
cat CQ8/layers_A.txt
echo "layers B:"
cat CQ8/layers_B.txt
```

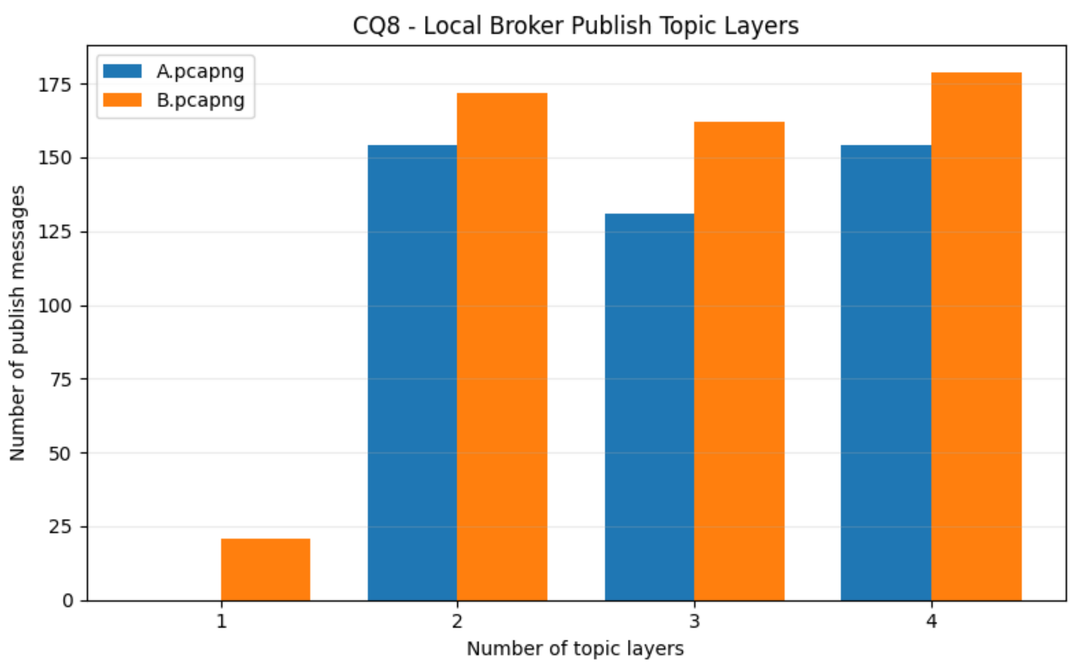
```
echo "layers B:"
cat CQ8/layers_B.txt
raw topics A count:
439 CQ8/topics_A_raw.txt
raw topics B count:
534 CQ8/topics_B_raw.txt
CQ8a total:
439 CQ8/topics_A.txt
CQ8b total:
534 CQ8/topics_B.txt
layers A:
```

	File: CQ8/layers_A.txt
1 ~	154 2
2 ~	131 3
3 ~	154 4

```
layers B:
```

	File: CQ8/layers_B.txt
1 ~	21 1
2 ~	172 2
3 ~	162 3
4 ~	179 4

```
→  challenge2 git:(master) ✗
```

Important note: some packets in B contain multiple MQTT publish entries in the same frame; counts above split comma-separated `mqtt.topic` entries so each publish message is counted separately.

In [16]:
```python
import matplotlib.pyplot as plt

# verified CQ8 distributions
layers_a = {2: 154, 3: 131, 4: 154}
layers_b = {1: 21, 2: 172, 3: 162, 4: 179}

all_layers = sorted(set(layers_a) | set(layers_b))
a_vals = [layers_a.get(k, 0) for k in all_layers]
b_vals = [layers_b.get(k, 0) for k in all_layers]

x = range(len(all_layers))
w = 0.38

plt.figure(figsize=(8, 5))
plt.bar([i - w / 2 for i in x], a_vals, width=w, label='A.pcapng')
plt.bar([i + w / 2 for i in x], b_vals, width=w, label='B.pcapng')
plt.xticks(list(x), all_layers)
plt.xlabel('Number of topic layers')
plt.ylabel('Number of publish messages')
plt.title('CQ8 - Local Broker Publish Topic Layers')
plt.legend()
plt.grid(axis='y', alpha=0.25)
plt.tight_layout()
plt.show()
```